

// strategic_intelligence_report.ipynb

EV Market

Analysis

Tesla Strategic Intelligence

PRESENTED BY

Alexxis Saucedo

COURSE

BUS 658 — Information & Data Systems

DATE

Spring 2026

Today's Presentation

01

Research Question & Scope

The strategic thesis

02

Analytical Framework & Data

Three pillars · 5 data sources

03

Market Context

Share decline vs. market growth

04

Pillar I — Chinese Competition

Zero presence, two data sources

05

Pillar II — Tariff Policy

May 2024 event study

06

Pillar III — FSD as Moat

Autonomy attention spikes

07

Demand Model & Findings

OLS regression · key results

08

Identification Challenges

Econometric limits

09

Competitor Analysis

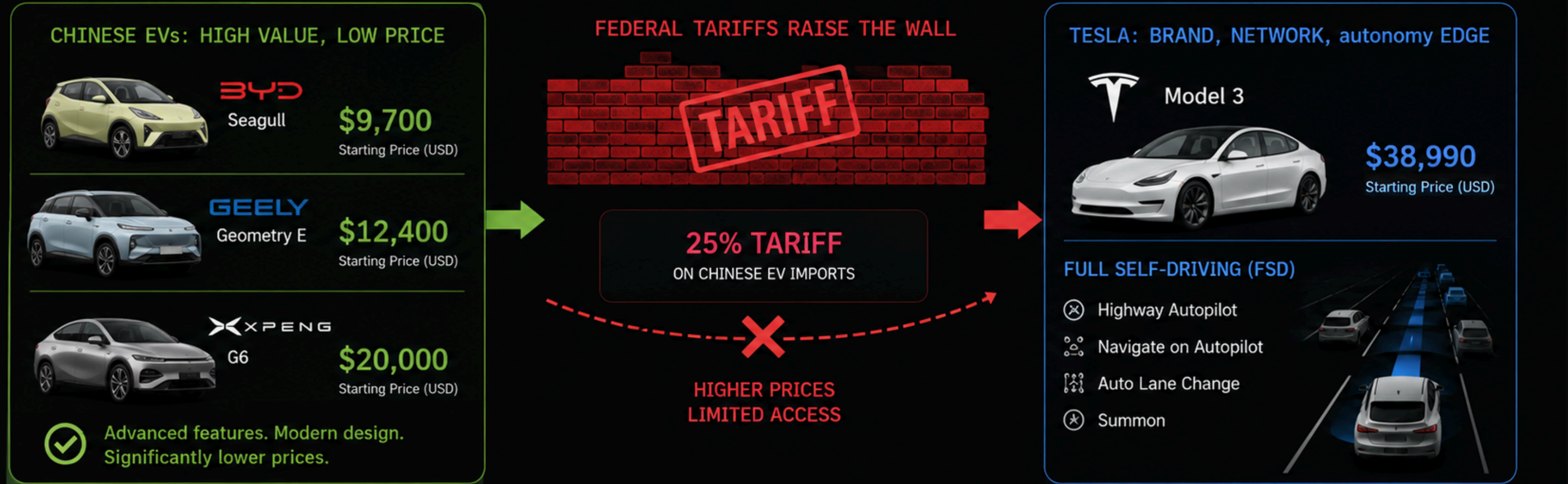
Ford + Hyundai/Kia

10

Strategic Synthesis & Recs

Three specific actions

How do Chinese EV competition, federal tariffs, and autonomous driving capabilities jointly shape US EV demand — and what does that suggest for Tesla's competitive position?



Methodology & Data Sources

DATASET	WHAT IT MEASURES	SOURCE	ROLE IN MODEL
EV Registrations	Adoption by state & make, 2020–2025	Atlas EV Hub	Dependent variable
Gas Prices	Monthly national avg, quarterly	EIA API	OLS predictor
Charging Infrastructure	Supercharger count + open dates	AFDC / NREL	Infrastructure density
Google Trends — 3 clusters	Brand attention proxy · FSD signals	pytrends	Competition index + events
Policy Event Dates	IRA (Aug 2022) · Tariff (May 2024)	Hardcoded	Treatment dummies

PANEL STRUCTURE

State × Quarter · 24 quarters · 298 obs

KEY EVENTS

IRA (EV tax credit): 2022-08-16 · FSD v12:
2024-03-01 · Tariff: 2024-05-14

METHOD

OLS · State FE + Quarter FE · R²=0.863

US EV Market Landscape

Within this 14-state sample, Tesla share declined from ~72% (early 2021) to ~55% (2025) while total BEV registrations grew ~10x.

Tesla is not losing customers in absolute terms — competitors appear to be capturing new buyers at a faster rate.

The IRA (Aug 2022) accelerated total BEV adoption, but Tesla's share continued declining through and after it.

~55%

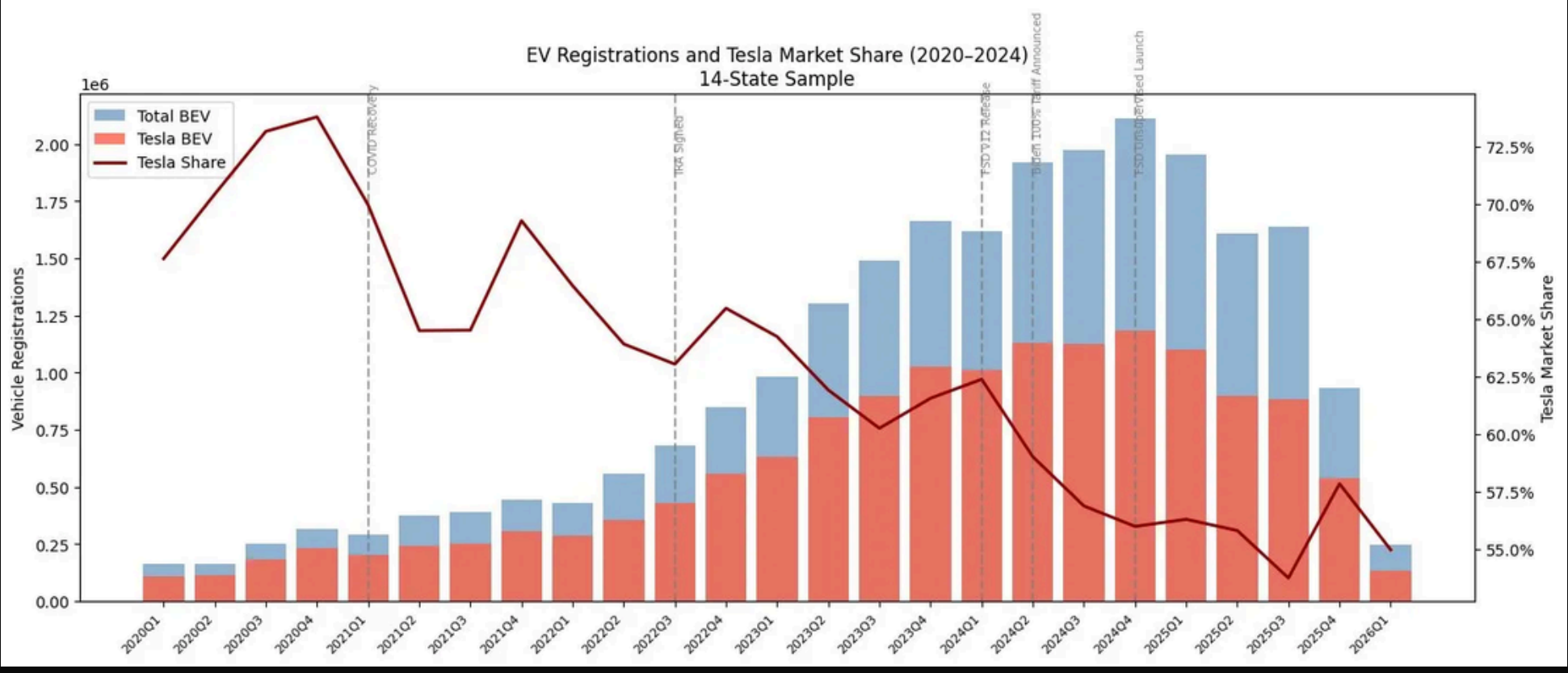
Tesla share, 2025

Down from ~72% in 2021

~10x

Total BEV growth

2020–2025, 14-state sample

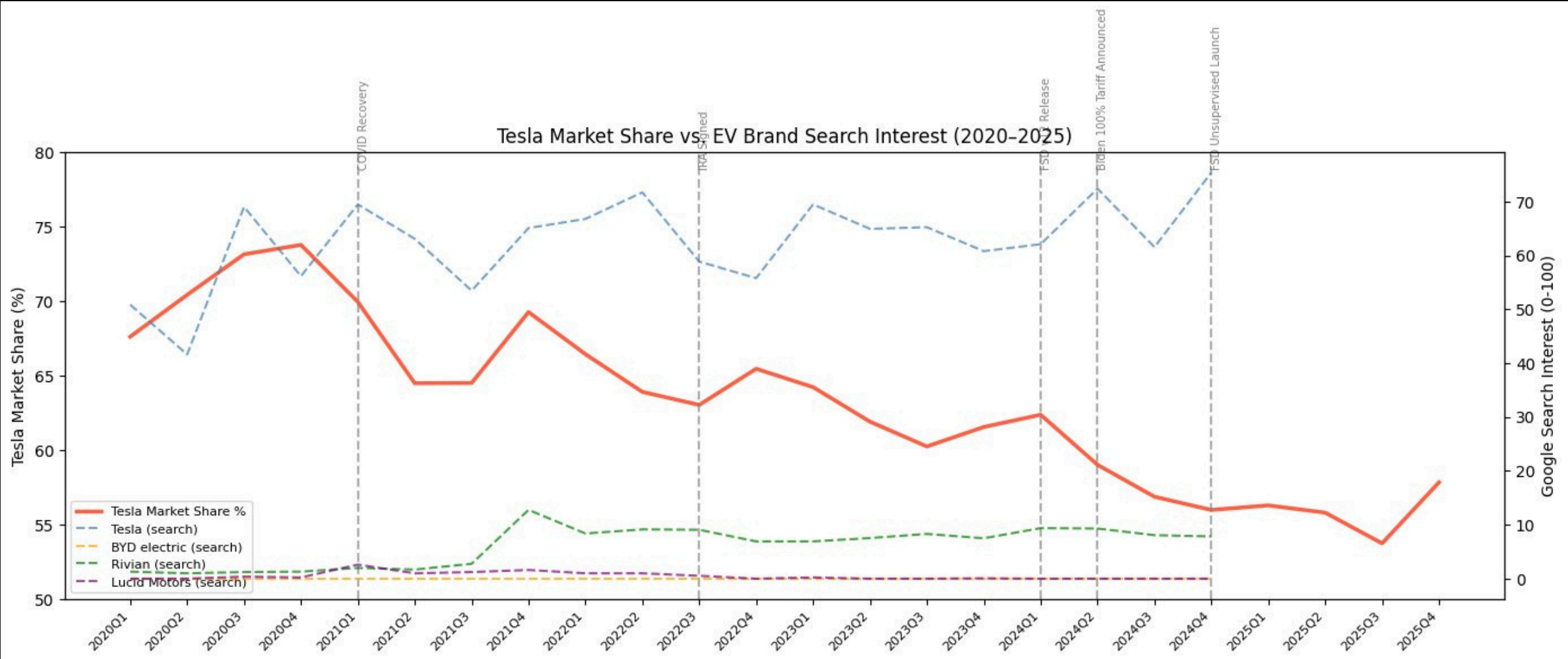


Chinese EVs Have Near-Zero US Consumer Awareness

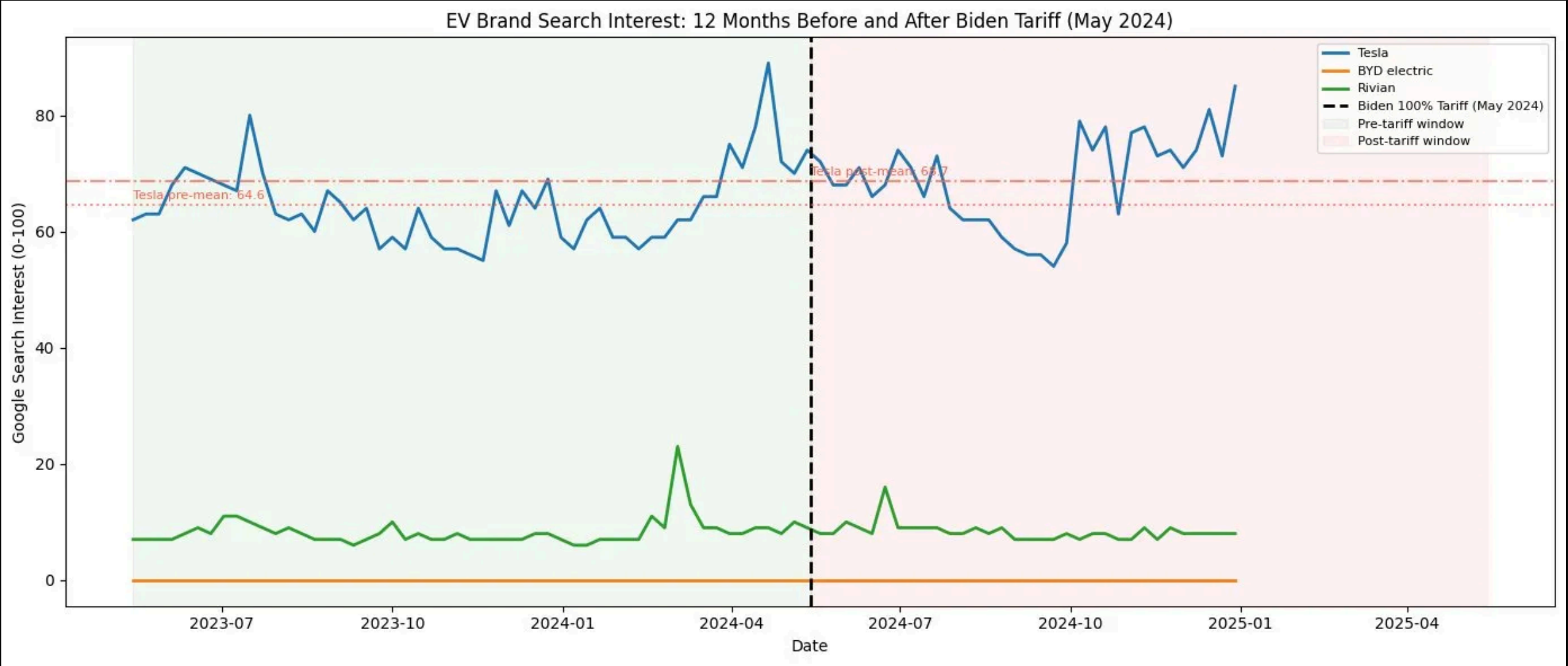
Google Trends (used here as a consumer attention proxy) shows near-zero Chinese EV awareness throughout 2020–2024. Registration data across 26M records confirms no meaningful Chinese EV passenger market presence.

KEY FINDING

BYD had 4,270 registrations vs. Tesla's 15.3 million — a 0.03% ratio consistent with zero meaningful passenger market presence. Combined with zero search interest across 262 weeks, two independent sources point to the same conclusion: the tariff is protecting Tesla from a threat that has not yet arrived.



May 14, 2024: The Tariff Inflection Point



WHAT THE DATA SHOWS

Tesla search interest: pre-tariff mean 64.6 → post-tariff mean 68.7 (+6.3%). BYD remained flat at zero throughout the event window.

IMPORTANT CAVEAT

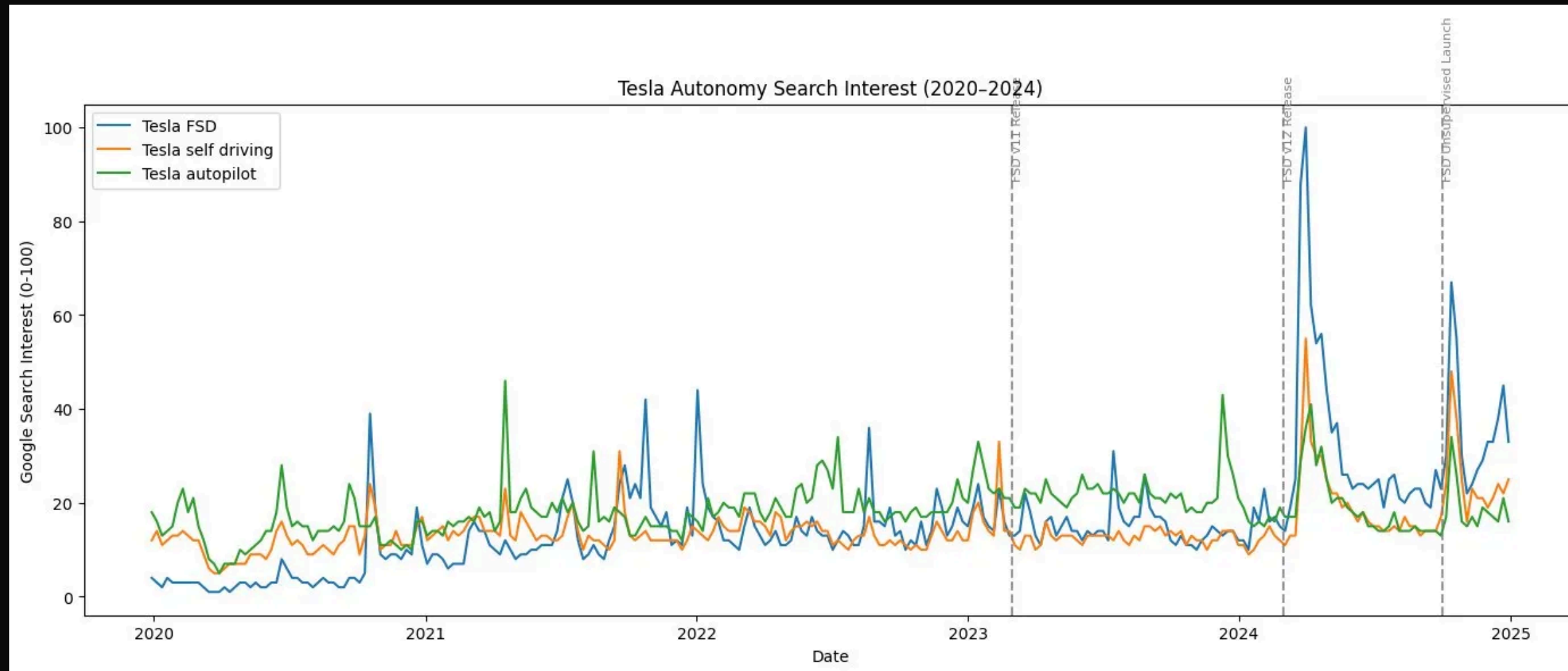
Attribution is difficult — the tariff coincided with Rivian's R2 launch (March 2024) and a broader macro EV slowdown. Isolating the tariff effect is not possible with this design.

REGRESSION RESULT

Post-tariff, within-state Tesla share fell -8.9 pp (coef: -0.089 , $p=0.040$), consistent with competitive erosion continuing despite the tariff shield.

[Source: Google Trends pytrends • 12-month event window • attention proxy only]

FSD: Strongest Observed Consumer Attention Signal in This Dataset



[Source: Google Trends pytrends • attention proxy • conversion to sales not measured]

+166%

FSD attention spike at v12

Pre: mean 12.5 → Post: mean 33.3

Among all signals observed in this dataset, autonomy-related events generated the largest consumer attention response. FSD v12 (March 2024) hit the index maximum of 100 — the single largest spike across all keywords and time periods.

CAVEAT

Search spikes may reflect media attention rather than purchase intent. Conversion from attention to sales is not observable in this data.

OLS Regression: What Drives Tesla's Market Share?

PREDICTOR	COEFFICIENT	DIRECTION	P-VALUE	STATUS
Post-Tariff (2024Q2+)	-0.089	↓ Negative	0.040 **	SIGNIFICANT
Gas Price (\$/gal)	-0.099	↓ Negative	0.018 *	SIGNIFICANT
Post-IRA (2022Q3+)	-0.059	↓ Negative	0.009 **	SIGNIFICANT
Rivian Search Interest	+0.241	↑ Positive	0.152	not sig.

0.863

R-squared

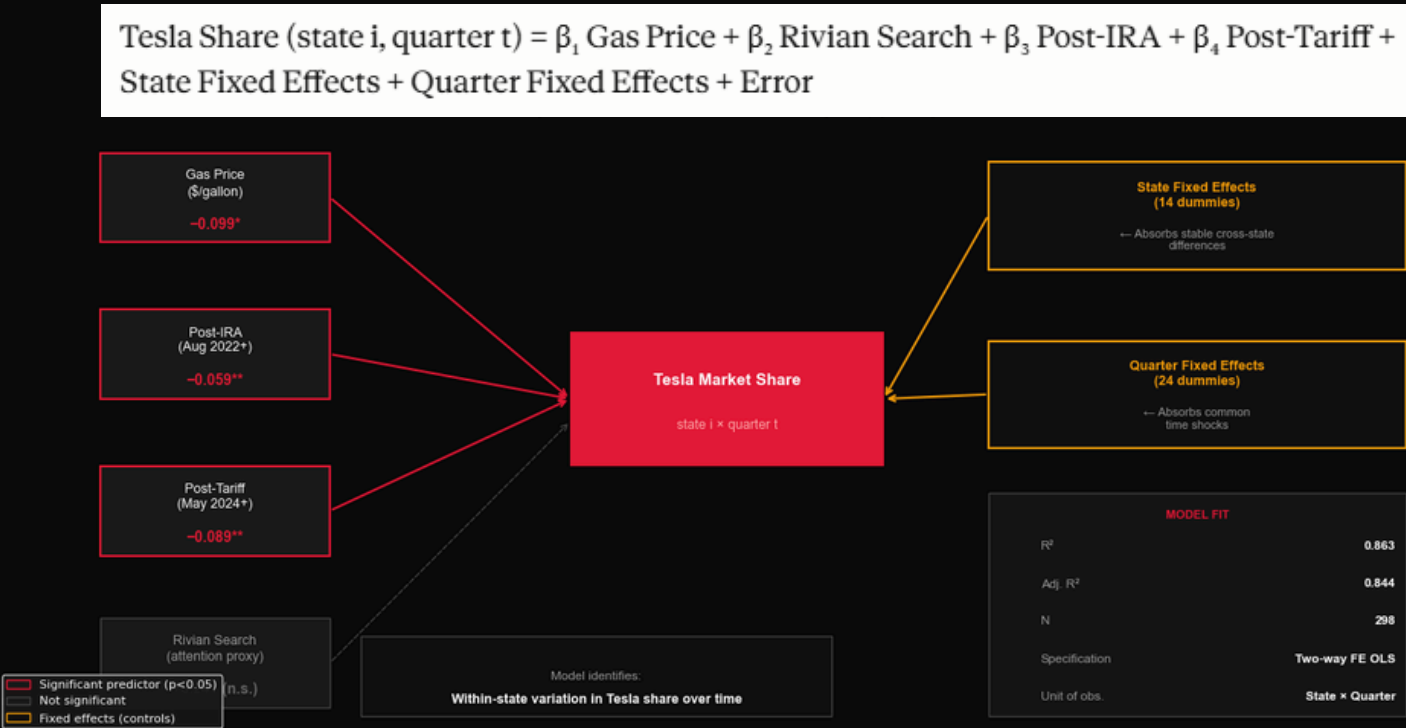
Two-way FE model

N = 298

state-quarter obs.

KEY FINDING

All three policy variables show negative significant associations. Results are consistent with competitive erosion continuing through both policy periods.



Econometric Limits of This Analysis

Results should be interpreted as directional strategic signals, not causal estimates. The following challenges apply:

PROXY VALIDITY

Google Trends measures consumer attention, not purchases. A search spike for 'Tesla FSD' may reflect media coverage rather than conversion intent. The relationship between search interest and actual demand is assumed, not measured.

CONFOUNDED TIMING

The May 2024 tariff announcement overlaps with Rivian's R2 launch (March 2024) and a broader macro EV slowdown. Isolating the tariff effect from concurrent events is not possible with this design.

REVERSE CAUSALITY

Higher Tesla market share could drive more Tesla search interest, not the other way around. The direction of causality between search signals and market outcomes is not identified.

OMITTED VARIABLES

Interest rates, EV financing costs, charging reliability, brand sentiment (including Musk-related), and state-level incentives beyond federal policy are not controlled for and may confound estimates.

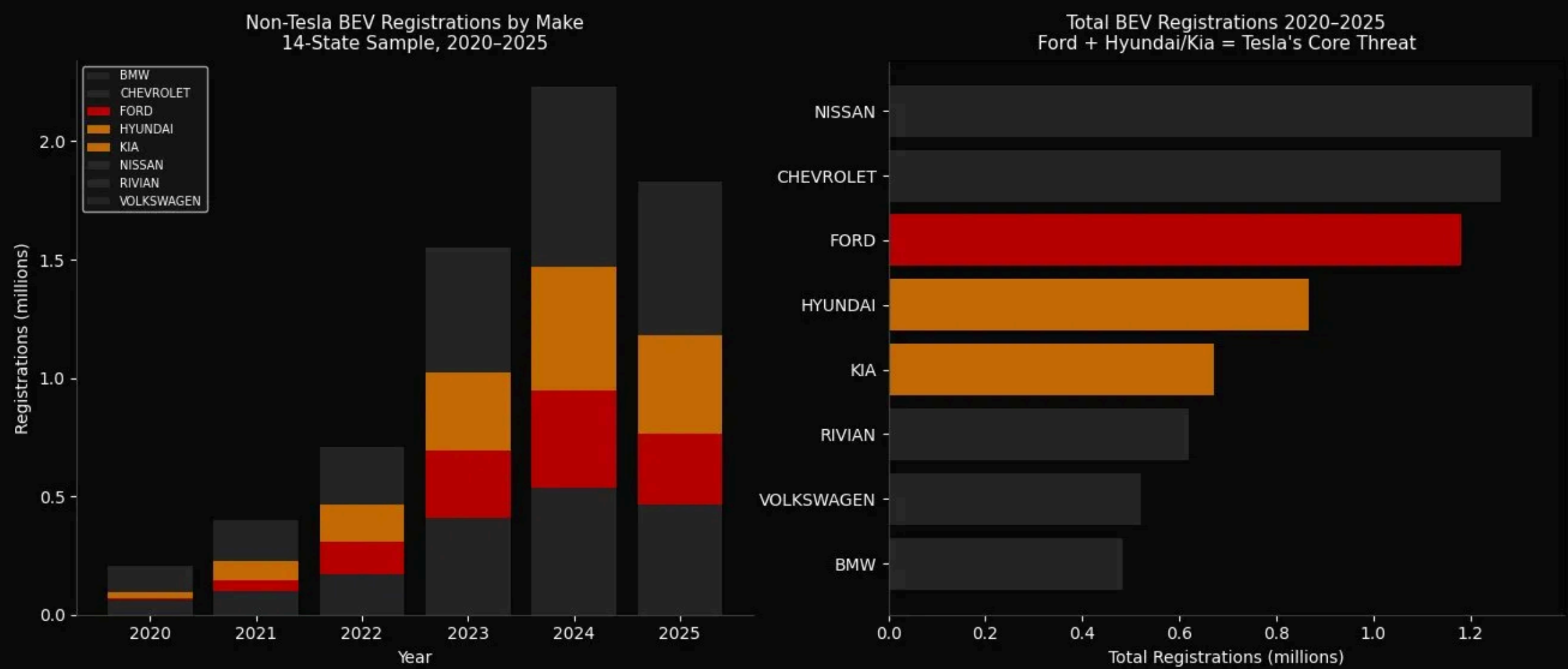
EXTERNAL VALIDITY

14-state sample excludes California (~35-40% of US EV sales). California was tested as a robustness check and showed consistent directional findings but was excluded due to annual vs. quarterly data mismatch.

DATA STRUCTURE

Google Trends is a relative index, not absolute search volume. Cross-pull comparisons are directional only. Gas price data required interpolation after mid-2021.

Tesla's Primary Domestic Competitors: Ford and Hyundai/Kia



[Source: Atlas EV Hub • 14 states • 2020-2025 • Nissan/Chevrolet volume includes legacy pre-2020 Leaf/Bolt registrations]

STRATEGIC IMPLICATION

FORD

Truck segment — F-150 Lightning targets buyers Tesla has no product for. Accelerate Cybertruck and a pickup-class vehicle.

HYUNDAI + KIA

Sub-\$40k sedan segment — EV6 and Ioniq 6 directly compete with Model 3 at lower price points. IRA-eligible buyers are choosing these over Tesla.

THE GAP

9 out of 10 new EV buyers in this sample went to non-Tesla brands. Tesla needs a product response in both segments — not just price cuts on existing models.

Three Data-Grounded Recommendations

01

Address Ford + Hyundai/Kia — Not Chinese EVs

Registration data shows 9 out of 10 new EV buyers in this sample went to non-Tesla brands.

Ford's F-150 Lightning wins the truck segment — Tesla has no direct response. Accelerate Cybertruck ramp and a pickup-class product.

Hyundai/Kia EV6 and Ioniq 6 win the sub-\$40k sedan segment. The IRA-eligible income brackets are buying these, not Model 3. Tesla needs a product response, not just a price cut.

Monetize the Supercharger network — now open to all EVs, it is an underexploited revenue stream.

02

Treat FSD Releases as Demand Events — Price Accordingly

Ford has no autonomy story at scale. Use this.

Coordinate FSD releases with regional availability expansions and sales pushes to convert attention into orders.

Price FSD as a recurring subscription — software revenue at scale is a margin story no traditional automaker can replicate.

Pursue regulatory approval in every remaining state — each state without FSD Unsupervised is a market where the attention advantage cannot convert.

03

Price Floor Preparation

BYD Seagull starts at ~\$10k in China. At 100% tariff → \$20k landed. At 50% → \$15k. At 25% → \$12.5k. Tesla has no product at any of those price points.

Accelerate unboxed manufacturing and Gigafactory cost reduction now — so Tesla has a viable response at multiple tariff scenarios, not just the current one.

The \$25k vehicle platform is a tariff hedge, not just a market expansion play. Every month it is delayed is a month of vulnerability if the tariff environment shifts.

The window to build that cost structure is while Chinese brands are locked out — not after they arrive.

The Window Is Real. The Clock Is Running.

Three findings, one strategic conclusion: the evidence is consistent with the competitive narrative being more nuanced than policy discourse suggests. The real threat is domestic, the tariff appears preemptive, and FSD is the strongest observed attention signal in this dataset.

- Chinese EV awareness: near-zero in US data. BYD: 0 search weeks out of 262. Zero passenger registrations across 14 states.
- Tariff: evidence is consistent with preemptive protection. Share continued declining post-tariff (−8.9 pp, $p=0.040$). Causal attribution is limited.
- **FSD: +166% attention spike at v12 — the strongest observed signal in this dataset. Primary domestic competitors are Ford and Hyundai/Kia.**

DELIVERABLES

01 — Jupyter Notebooks (data collection, EDA, model)

02 — Panel dataset (14 states × 24 quarters, 298 obs)

03 — OLS two-way FE regression ($R^2=0.863$)

04 — Written report + strategic implications

05 — This slide deck